Office of Research Assurance

Human Subject Research Protections Standard Operating Procedures (SOP)

SOP- 720: : Return of Research Results

Number	Effective Date	Version Number	Previous Versions
SOP-720	5/1/2024	v.1	N/A

Purpose

SOP-720 defines the NASA IRB policy and processes with respect to planning for and handling the return of research results to individual study participants. Please note this SOP does not change any other NASA medical policies or procedures for medical treatment or care of crewmembers.

Individual research results are information linked to the identity of a research participant, that both pertains to that research participant and is **the focus of the research study.** Examples include an individual's ability to withstand a given gravitational force and still perform a task, psychological findings, or the disclosure of the study arm to which a participant was randomized. Individual research results can be clinical/medical or social behavioral.

Individual research results are broken down into three categories, which are based on when the information was created:

1. Baseline findings – results collected at the beginning of the individual's participation to determine if the participant qualifies for the research (screening to determine inclusion/exclusion criteria)
2. In-study findings – results generated during the progress of the research
3. End-of-study findings – results collected, collated, and/or interpreted after the study has ended and data analysis for all study participants is complete

The return of individual research results should be viewed as the "offering" of individual research results. Participants should be given the opportunity to decide whether or not to receive the results. Legal reporting requirements, such as mandated reporting of communicable diseases, abuses, and intent to harm self or others are an exception.

Institutional Review Boards (IRBs) have a responsibility to evaluate the return of individual research results to participants as such return is supported by the Belmont principles of

respect for persons and beneficence. In regard to respect for persons, researchers have a duty toward research participants that includes identifying, verifying, and communicating individual research results of clinical significance, even though researchers are not serving in a clinical capacity during research. Disclosure of individual research results also builds a sense of partnership in the research process, leading to greater public trust and willingness to participate in research. Beneficence provides a directive to avoid harm to participants. To the extent that the return of individual research results to participants could lead to harm or discomfort, whether physical, psychological, financial, reputational, or social, particularly with no offset of direct benefit, there must be a consideration of whether it is appropriate to return such findings. Although the disclosure of individual research results may prove valuable to participants and their families in regard to their health or welfare, participants may misuse or misunderstand individual research results provided to them and take inappropriate medical or personal action. Participants may also misconstrue the purpose of the research, failing to distinguish between research and clinical care. Lastly, the maintenance of participant contact information increases risks to participants for data breaches and unauthorized disclosure.

It is possible for research to be conducted respectfully without the return of individual research results. If participants are made aware during the informed consent process that individual research results will not be returned, and they choose to participate in the study, their autonomy has been respected. Evaluation of the benefit of returning individual research results should also consider:

- The potential to cause bias in the study (mainly with respect to timing of disclosure)
- Whether the meaning and significance of the research results are known
- If results have low importance or insignificance
- Lack of validity
- Lack of actionability
- Administration and/or financial burden to the investigator, institution, and other parties

Additionally, in some studies, it may be impossible to provide individual research results because the participants are de-identified or anonymous (e.g., certain tissue banking activities, research conducted under a waiver of informed consent, exempt studies, and cluster randomized studies).

IRBs must consider the risks and benefits of disclosing or returning individual research results during the design, conduct, and oversight of research acknowledging the rebuttals but presuming disclosure. The return of individual research results is permissible, but not mandatory and the decision to disclose research results to individual participants will be made on a case-by-case basis.

When determining whether to return individual research results, the more likely the results are validated and actionable, the more justification for disclosing (returning) them to participants. Actionable can mean either clinically actionable or personally actionable. Validated usually means the test or diagnosis is conducted using devices or assays that have Food and Drug Administration (FDA) approval or certification under Clinical Laboratory

Improvement Amendments of 1988 (CLIA)[1] from the Centers for Medicare and Medicaid Services (CMS). Generally, any individual research result that reveals a condition or information likely to be life-threatening, a grave condition that can be avoided or ameliorated, or information that affects reproductive decision-making should be reported to participants in a timely manner (adapted from Wolf et al., 2008).

	Validated	**Not Validated**
Actionable	Discovered when reviewing results of standard clinical tests (e.g., pregnancy tests, EKGs, MRIs, chest X-rays) • Strong presumption to disclose	Fact dependent and when able, should follow up with validated testing to help provide certainty • May disclose
Not Actionable	Fact dependent and the concept of personal utility should be considered • Disclosure recommended	Experimental results (e.g., genetic variant not known to be associated with a disease) • Not a strong presumption to disclose

Policy

Regardless of the return of individual research results happening at baseline, in-study, or end-of-study:

1. For studies involving NASA astronauts as participants, the Protocol Compliance Officer or Alternate Compliance Officer, in consultation with the NASA IRB if deemed necessary, will determine if individual research results are actionable and/or validated. For studies not involving NASA astronauts as participants, the NASA IRB Chair/designee, in consultation with the NASA IRB or any other experts if deemed necessary, will determine if individual research results are actionable and/or validated.
2. For studies involving NASA astronauts as participants, the Protocol Compliance Officer or Alternate Compliance Officer, in consultation with the NASA IRB if deemed necessary, will determine if individual research results reveal a condition or

[1] Research testing where patient-specific results are reported from the laboratory, and those results will be or could be used 'for the diagnosis, prevention, or treatment of any disease or impairment of, or the assessment of the health of, human beings' is presumed to be subject to CLIA absent evidence to the contrary.

information likely to be life-threatening, a grave condition that can be avoided or ameliorated, or information that affects reproductive decision-making. For studies not involving NASA astronauts as participants, the NASA IRB Chair/designee, in consultation with the NASA IRB or any other experts if deemed necessary, will determine if individual research results reveal a condition or information likely to be life-threatening, a grave condition that can be avoided or ameliorated, or information that affects reproductive decision-making.

3. When prospective consent has not been solicited for the return of individual research results:
    a. For studies involving NASA astronauts as participants, the Protocol Compliance Officer or alternate, in consultation with the NASA IRB if deemed necessary, will determine if individual research results will be disclosed (or returned) to the participant(s).
    b. For studies not involving NASA astronauts as participants, the NASA IRB Chair/designee, in consultation the NASA IRB or any other experts if deemed necessary, will determine if individual research results will be disclosed (or returned) to the participant(s).
4. When warranted, individual research results will be returned to participants, or their legally authorized representative, directly (this may be in oral or written form).
5. For studies involving NASA astronauts as participants, the Protocol Compliance Officer or Alternate Compliance Officer will disclose (return) or is to be involved in the disclosure (return) of individual research results. Individual research results that are determined by the Protocol Compliance Officer or alternate to be a real or potential risk to the mission will require timely notification to the NASA astronaut and his/she/their assigned flight surgeon. After this, if necessary, individual research results may be placed in the participant's medical record.
6. For studies not involving NASA astronauts as participants, where possible, an individual with a clinical relationship with the research participant will disclose (return) or is to be involved with the disclosure (return) of individual research results. Individual research results might also be provided to the participant's health care provider or be placed in his/her/their medical record.
7. Principal Investigators (PIs) are not to disclose individual research results to participants unless otherwise deemed acceptable or necessary by the NASA IRB.
8. If a researcher has an ethical objection to allowing participants to opt out of receiving clinically significant, actionable, validated, lifesaving individual research results, he/she/they need not enroll these individuals in the research study and they must outline this exclusion criterion in the protocol and informed consent form.
9. If a researcher discovers a clinically significant, actionable, validated, lifesaving individual research result for a participant who has opted out of receiving individual research results, the investigator should seek advice from the NASA IRB. Generally, the PI should reconfirm with the participant who indicated refusal that he/she/they are

electing to decline information on all individual research results, even those with high health importance that can be treated.

10. For studies in which a Reliance Acknowledgment is executed:
    a. When NASA is the IRB of Record, IRB protocols will adhere to this policy
    b. When NASA has ceded review to another institution, the IRB of Record's institutional policies govern, although the NASA IRB reserves the right to make recommendations on a protocol-specific basis

Responsibilities

Principal Investigator

1. Whether or not the researcher is also the treating physician, if the researcher or a member of the research team identifies individual research results of potential concern, it is the PI's duty to handle these findings responsibly and promptly.
2. At the time of initial IRB review:
    a. Researchers are encouraged to consult with an expert (e.g., radiologist, clinical geneticist, etc.) to determine which potential individual research results may be of clinical significance during protocol development.
    b. The researcher must describe the return of individual research results in the protocol and consent form.
        i. The protocol must outline:
            1. The basic information regarding the program for the release of data, including what information will be released, who will do so, and when the disclosure is expected to take place; and
            2. The process for keeping participant contact information readily available after the study is finished, specific if, how, and for how long such information is to be retained
        ii. The informed consent form must include:
            1. The intent to provide individual research results;
            2. The basic information regarding the program for the return of results, including a basic description of the information to be returned, who will provide the information, and when the disclosure will take place;
            3. The potential risks/benefits of disclosure;

4. A statement that findings during a research study cannot be assumed to be of clinical quality and there, cannot substitute a clinical evaluation; and
5. A statement whether findings will be placed in the participant's medical record

Note: Participants should be told if return of individual research results will or will not be possible when consenting to data use for future research.

c. Or, the researcher must outline that individual research results will not be returned to participants in the protocol and consent form.

Note: NASA IRB approval for returning results may be withheld in the absence of CLIA certifications where appropriate.

3. Post-IRB Approval and while the study is still open:
   a. Researchers are encouraged to consult with an expert (e.g., radiologist, clinical geneticist, etc.) to determine which potential individual research results may be of clinical significance. Consultation should not provide information that would disclose the participant's identity to protect the participant's privacy and confidentiality.
   b. If the researcher now wishes to return individual research results, and this was not previously outlined in the initial submission, he/she/they must:
      i. Submit a modification OR reportable new information (RNI) in the eIRB.
         1. The modification/RNI summary must outline:
            a. an opinion as to whether the individual research results are actionable and/or validated with supporting information;
            a. a recommendation for consultants where appropriate;
            b. the risks and benefits of disclosure;
            c. the disclosure plan – by whom and when; and
            d. if able, confirmation that the participants prospectively agreed to the disclosure
         2. If a modification, the protocol must be modified to outline:
            a. The basic information regarding the program for the release of data, including what information will be released, who will do so, and when the disclosure is expected to take place; and



b. The process for keeping participant contact information readily available after the study is finished, specific if, how, and for how long such information is to be retained

3. If a modification, the informed consent form must be modified to include:
    a. The intent to provide individual research results;
    b. The basic information regarding the program for the return of results, including a basic description of the information to be returned, who will provide the information, and when the disclosure will take place;
    c. The potential risks/benefits of disclosure;
    d. A statement that findings during a research study cannot be assumed to be of clinical quality and there, cannot substitute a clinical evaluation; and
    e. A statement whether findings will be placed in the participant's medical record
4. The modification should also outline the process to reconsent previously enrolled participants if applicable.

ii. Wait for direction from the NASA IRB regarding recommendations for next steps before taking action

Note: NASA IRB approval for returning results may be withheld in the absence of CLIA certifications where appropriate.

4. After study closure:
    a. Researchers are encouraged to consult with an expert (e.g., radiologist, clinical geneticist, etc.) to determine which potential individual research results may be of clinical significance. Consultation should not provide information that would disclose the participant's identity to protect the participant's privacy and confidentiality.
    b. If the researcher now wishes to return individual research results, and this was not previously outlined in the approved IRB protocol, he/she/they must:
        i. Submit reportable new information (RNI) in the eIRB outlining:
            1. an opinion as to whether the individual research results are actionable and/or validated with supporting information;
            2. a recommendation for consultants where appropriate;
            3. the risks and benefits of disclosure;
            4. the disclosure plan – by whom and when;

5. if able, confirmation that the participants prospectively agreed to the disclosure;
6. The basic information regarding the program for the release of data, including what information will be released, who will do so, and when the disclosure is expected to take place;
7. The process for keeping participant contact information readily available after the study is finished, specific if, how, and for how long such information is to be retained; and
8. Plan and process for informing/reconsenting enrolled participants of:
    a. The intent to provide individual research results;
    b. The basic information regarding the program for the return of results, including a basic description of the information to be returned, who will provide the information, and when the disclosure will take place;
    c. The potential risks/benefits of disclosure;
    d. A statement that findings during a research study cannot be assumed to be of clinical quality and there, cannot substitute a clinical evaluation; and
    e. A statement whether findings will be placed in the participant's medical record

ii. Wait for direction from the NASA IRB regarding recommendations for next steps before taking action

Note: NASA IRB approval for returning results may be withheld in the absence of CLIA certifications where appropriate.

NASA IRB

1. At the time of initial IRB review:
    a. The NASA IRB will review the plan to return individual research results and ensure information provided in the protocol and consent form are appropriate.
    b. The NASA IRB will determine whether the plan to return individual research results affects the approval criteria at 14CFR1230.111.
2. Post-IRB approval and while the study is still open:
    a. Review of the modification will determine whether the proposed change satisfies the approval criteria at 14CFR1230.111.



   b. Review of the modification will be expedited if the outlined change does not affect the risk to benefit ratio.
   c. The NASA IRB may request reconsent if the new information may affect one's willingness to continue participation.

3. After study closure:
   a. Although the activity is no longer human subjects research per 14CFR1230, the NASA IRB will review the reportable new information as risk to the participant's confidentiality or risk of misinterpretation/misunderstanding is possible.
   b. Decisions to re-open the study will be made by the NASA IRB Chair/delegate on a case-by-case basis.
4. Upon receipt of a modification or RNI in the eIRB outlining the return of individual research results, the NASA IRB Staff will:
   a. notify the NASA IRB Chair/designee for studies not involving NASA astronauts as participants to determine next steps; and/or
   b. notify the NASA IRB Chair/designee and the Protocol Compliance Officer for studies involving NASA astronauts as participants to determine next steps
5. The NASA IRB will provide consultation on individual research results as appropriate.
6. The NASA IRB will facilitate all communication of individual research results with the Protocol Compliance Officer when appropriate.

Definitions

1. Individual research results - information linked to the identity of a research participant, that both pertains to that research participant and is **the focus of the research study.**
2. Baseline findings – results collected at the beginning of the individual's participation to determine if the participant qualifies for the research (screening to determine inclusion/exclusion criteria)
3. In-study findings – results generated during the progress of the research
4. End-of-study findings – results collected, collated, and/or interpreted after the study has ended and data analysis for all study participants is complete
5. Actionable – ethically requiring action - can mean either clinically actionable or personally actionable
6. Validated - usually means the test or diagnosis is conducted using devices or assays that have Food and Drug Administration (FDA) approval or certification under Clinical Laboratory Improvement Amendments of 1988 (CLIA) from the Centers for Medicare and Medicaid Services (CMS)

7. Protocol Compliance Officer- a physician from NASA SD that is also a NASA IRB member. He/she/they shall have no conflict of interest with potential crewmembers.
8. Alternate Protocol Compliance Officer- an alternate physician from NASA SD, who is a NASA IRB member, that possesses no conflict of interest with potential crewmembers, and fulfills the responsibilities of the protocol compliance officer when he/she/they is/are unable to.

Acronyms

IRB – Institutional Review Board
PI – Principal Investigator
NASA – National Aeronautics and Space Administration

References

1. Belmont Report
2. Clinical Laboratory Improvement Amendments of 1988
3. Wolf et al. (2008). Managing Incidental Findings in Human Subjects Research. *Journal of Law, Medicine & Ethics, 36*(2), 219-211.
4. Secretary's Advisory Committee on Human Research Protections (SACHRP) Recommendation – Attachment B – Return of Individual Research Results

SOP Role	Name	Date
Author	J. Kisenwether	08April2023
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